RUSTAM LATYPOV

Sublinear Algorithms Group
Aalto University, Finland



I'm a postdoc in the group of Jara Uitto at Aalto University with research and teaching experience in algorithm design and combinatorics. My research interests lie in the mathematical foundations of distributed and parallel computing. I specialize in fundamental graph problems in modern models of computing. During my free time I climb rocks.

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Postdoctoral researcher in the Sublinear Algorithms Group, Aalto University

EDUCATION

+ **Doctor of Science** · Computer Science — Aalto University, 2026
  Advisor Prof. Jara Uitto · Opponent Prof. Artur Czumaj — phd-thesis
+ **Master of Science** · Mathematics and Operations Research — Aalto University, 2021
  GPA 4.9/5 — rustamlatypov/masters-thesis
+ **Bachelor of Science** · Engineering Physics and Mathematics — Aalto University, 2019
  GPA 4.9/5 — rustamlatypov/bachelors-thesis

AWARDS

★ Awarded 5000€ – Encouragement grant, The Finnish Foundation for Technology Promotion [link]
★ Awarded 5000€ – Nokia Scholarship, Nokia Foundation [link]
★ Received 24-month funding for my doctoral studies (out of 20 applicants) – Dept. of CS, Aalto [link]
★ Awarded 1000€ for academic success in mathematics – Professor E. J. Nyström Fund [link]
★ Awarded 3 × 500€ for general academic success – School of Science, Aalto [link]

PUBLICATIONS (authors are ordered alphabetically, as is standard in the field)

11. Near-Optimal Distributed 2-Ruling Sets on Graphs with Low Arboricity [doi, arXiv]
Malte Baumecker, Rustam Latypov, Yannic Maus, Jara Uitto
International Symposium on Distributed Computing (DISC) 2026

10. Massively parallel algorithms for sparse graphs [isbn]
Rustam Latypov
Aalto University publication series Doctoral Theses 82/2026

9. Near-Optimal Directed Low-Diameter Decompositions [doi, arXiv]
Karl Bringmann, Nick Fischer, Bernhard Haeupler, Rustam Latypov
International Colloquium on Automata, Languages, and Programming (ICALP) 2025

8. A Simple Parallel Algorithm with Near-Linear Work for Negative-Weight Single-Source Shortest Paths [doi, arXiv]
Nick Fischer, Bernhard Haeupler, Rustam Latypov, Antti Roeyskoe, Aurelio Sulser
SIAM Symposium on Simplicity in Algorithms (SOSA) 2025

7. Adaptive Massively Parallel Coloring in Sparse Graphs [doi, arXiv]
Rustam Latypov, Yannic Maus, Shreyas Pai, Jara Uitto
ACM Symposium on Principles of Distributed Computing (PODC) 2024

6. Conditionally Optimal Parallel Coloring of Forests [doi, arXiv]
Christoph Grunau, Rustam Latypov, Yannic Maus, Shreyas Pai, Jara Uitto
International Symposium on Distributed Computing (DISC) 2023

5. Adaptive Massively Parallel Connectivity in Optimal Space [video, doi, arXiv]
Rustam Latypov, Jakub Lacki, Yannic Maus, Jara Uitto
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2023

4. Fast Dynamic Programming in Trees in the MPC Model [doi, arXiv]
Chetan Gupta, Rustam Latypov, Yannic Maus, Shreyas Pai, Simo Särkkä, Jan Studený, Jukka Suomela, Jara Uitto, Hossein Vahidi
ACM Symposium on Parallelism in Algorithms and Architectures (SPAA) 2023

3. Optimal Deterministic Massively Parallel Connectivity on Forests [doi, arXiv]
Alkida Balliu, Rustam Latypov, Yannic Maus, Dennis Olivetti, Jara Uitto
ACM-SIAM Symposium on Discrete Algorithms (SODA) 2023

2. Exponential Speedup Over Locality in MPC with Optimal Memory [journal, doi, arXiv]
Alkida Balliu, Sebastian Brandt, Manuela Fischer, Rustam Latypov, Yannic Maus, Dennis Olivetti, Jara Uitto
International Symposium on Distributed Computing (DISC) 2022

1. Memory Efficient Massively Parallel Algorithms for LCL Problems on Trees [video, doi, arXiv]
Sebastian Brandt, Rustam Latypov, Jara Uitto
International Symposium on Distributed Computing (DISC) 2021

ACADEMIC ACTIVITY

Reviewed for Conferences: FOCS'26, PODC'26, STOC'26, SODA'26, FOCS'25, PODC'25, DISC'24, ICALP'24, ITCS'24, SWAT'24, PODC'23, DISC'23, PODC'22, DISC'22, DISC'21, OPODIS'20; and Journals: TCS

Teaching at Aalto University

- Advanced Course in Algorithms – Fall 2022, 2023 (Head TA) [link]
- Principles of Algorithmic Techniques – Fall 2021, 2023, 2025 (TA / Head TA) [link]

Supervised an MSc thesis

- Hannes Sederholm, 2022 – *Distributed Drawing of Planar Graphs in the CONGEST model* [link]

PROGRAMMING PROJECTS – github.com/rustamlatypov

* **Parallel matrix multiplication** C++
  Parallel (CPU) matrix multiplication achieving 500-fold speedup w.r.t. sequential
* **Tile-matching game** C++
  Tournament grade Tetris and Pentis with controls in accordance with the Super Rotation System
* **Parallel radix sort** Scala
  Sequential and parallel (CPU) LSD radix sorts achieving 5- and 15-fold speedups w.r.t. scala.quickSort
* **Inverse problem for resistor networks** MATLAB
  Solving non-linear, ill-posed inverse problems for resistor networks both symbolically and numerically
* **Machine learning classifier for music genres** Python
  Solving a skewed, multiclass music genre classification problem with supervised PCA and SVM
* **Parallel password cracker** Python
  A command-line tool for cracking passwords in parallel (CPU) using dictionary and hybrid attacks

WORK EXPERIENCE

- **Postdoctoral researcher** · Aalto University May 2026 –
  Sublinear Algorithms Group
- **Doctoral researcher** · Aalto University Jun 2021 – April 2026
  Worked on the mathematical foundations of distributed and parallel computing. My work mainly focused on graph coloring and connectivity problems in different parallel models of computation.
- **Research assistant** · Aalto University Mar 2020 – May 2021
  Worked on distributed graph algorithms in the Massively Parallel Computation model. Developed and published a state-of-the-art graph coloring algorithm (Master's thesis).
- **Giant Leap Intern** · Vaisala Oyj Jun – Aug 2019
  Developed software that successfully forecasted test failures in large scale radiosonde production using data mining, feature engineering and machine learning (XGBoost, Python).
- **Research assistant** · Aalto University Jun – Aug 2018
  Studied ill-posed non-linear inverse problems for resistor networks both symbolically and numerically (Bachelor's thesis).